CHAPTER 8: Number Systems

Question 1: Multiply $35_6 \times 15_6$ directly in base 6

Calculations:

We multiply 35_6 by 15_6:

3 5 (base 6)
x 1 5 (base 6)

Step 1: Multiply 35_6 by 5_6

$5_6 \times 5_6 = 25_{10}$.
$25_{10} = (4 \times 6) + 1 = 41_6$.
Write down 1, carry 4.
$5_6 \times 3_6 = 15_{10}$.
Add carry 4: $15 + 4 = 19_{10}$.
$19_{10} = (3 \times 6) + 1 = 31_6$.
Write down 31.
First Partial Product: 311_6.

Step 2: Multiply 35_6 by 1_6 (shifted left by 1 digit)

$1_6 \times 35_6 = 35_6$.
~~Sicond~~ Second Partial Product: 350_6.

Step 3: Add the partial products in base 6

3 1 1 (base 6)
+ 3 5 0 (base 6)

Units column: $1 + 0 = 1$.
Sixes column: $1 + 5 = 6_{10} = 10_6$. Write down 0, carry 1.
Thirty-sixes column: $3 + 3 + 1$ (carry) $= 7_{10} = 11_6$. Write down 11.

$35_6 \times 15_6 = 1101_6$

(Verification in base 10: $35_6 = 23_{10}$, $15_6 = 11_{10}$. $23 \times 11 = 253_{10}$.
Converting $1101_6 = 1(216) + 1(36) + 0(6) + 1(1) = 253_{10}$. ~~Matchfes~~ Matches perfectly!)

Which was harder, this or 23×11 in base 10? Why?

Multiplying in base 6 felt noticeably harder. Since childhood, our minds have internalized the base 10 addition and multiplication tables into subconscious muscle memory. When multiplying in base 6, every single operation forces you to break automatic reflexes: a product of $\$25\$$ cannot simply be written as 25; you have to consciously decompose it into groups of 6 ($\$4 \times 6 + 1 = 41_6\$$). This mental translation layer makes even simple arithmetic feel foreign and demanding.

Question 2: Impact of Video on Understanding Computers

Watching the physical mechanics of computing demystifies the black box. It reinforces that computers are not magical abstract brains—they are deterministic physical systems built out of simple logic gates switching voltages between high and low. Understanding how binary representations and clock cycles translate logic into arithmetic bridges the gap between high-level programming and physical hardware.